
AAK Academy Management System

Academic and Administrative Automation Platform

Project Proposal Report

Department of Computer Science

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Contents

1	Introduction	2
2	Problem Statement	2
3	Objectives	3
4	Scope	4
4.1	Included Features	4
4.2	Excluded from Initial Release	4
5	Data Collection	5
5.1	Primary Sources	5
5.2	Secondary Sources	5
5.3	Data Entities Managed by the System	6
6	Technology Stack	6
7	System Functionality	7
8	User Classes	8
9	Task Division and Collaboration	9
9.1	Salman Safdar (Group Lead)	9
9.2	Muhammad Haris	10
9.3	Farwa Imran	10
10	Conclusion	10
A	AI Usage Disclosure	11

1. Introduction

The **AAK Academy Management System (AAMS)** is a web-based academic and administrative automation platform developed specifically for AAK Academy (Ammar Ahmad Khan Academy), an educational institution in Pakistan offering Matric (9th–10th grade) and Intermediate (11th–12th grade) programs.

In today's digital era, educational institutions face mounting pressure to modernize their operations. Traditional paper-based processes for student enrollment, fee management, assignment distribution, and performance tracking are slow, error-prone, and resource-intensive. There is a growing global need for integrated digital platforms that can bring together all aspects of academy management under a single, accessible interface.

AAMS addresses this need by providing a centralized, role-driven platform that connects students, teachers, and administrators in a seamless digital environment. The system is particularly relevant in the Pakistani educational context, where academies and coaching centers operate with limited administrative resources yet must deliver high-quality education. By automating key processes such as subject selection with discount calculation, assignment management, quiz administration, and live class scheduling, AAMS significantly reduces manual workload while improving transparency and academic outcomes.

The real-world relevance of this system extends beyond convenience: it enables data-driven decision-making, ensures financial accuracy in fee processing, and provides students with 24/7 access to learning materials – factors that directly contribute to improved educational quality and institutional efficiency.

2. Problem Statement

AAK Academy currently relies on manual, paper-based workflows to manage its academic and administrative operations. This gives rise to several critical problems:

- **Enrollment Inefficiency:** Student registration and subject selection are handled manually, causing delays and frequent data entry errors.
- **Fee Calculation Errors:** Manual computation of fees and discount application (10% for 3 subjects; 20% for 6 subjects) leads to financial inaccuracies and student disputes.
- **Fragmented Communication:** Assignment distribution, deadline notifications, and announcements rely on informal channels (WhatsApp, verbal instruction), resulting in missed information.

- **No Centralized Performance Data:** Student grades and progress are recorded in isolated spreadsheets or notebooks, making it impossible to generate meaningful analytics or early intervention signals.
- **Poor Resource Accessibility:** Learning materials, lecture notes, and past papers are shared inconsistently, limiting student access to study resources outside class hours.
- **Administrative Overload:** Administrators spend significant time on repetitive tasks such as generating invoices, tracking payments, and maintaining student records manually.

There is therefore a clear and urgent need for an integrated digital management system that automates these processes, eliminates data silos, and provides a unified platform for all stakeholders.

3. Objectives

The AAK Academy Management System is designed to achieve the following measurable objectives:

1. **Automate Student Enrollment:** Enable online registration and subject selection, reducing enrollment processing time by at least 70% compared to manual methods.
2. **Eliminate Fee Calculation Errors:** Implement a real-time, rule-based fee calculation engine that automatically applies discounts (10% for 3 subjects; 20% for 6 subjects) with zero manual intervention.
3. **Digitize Academic Content Delivery:** Provide a structured repository for lecture notes, video tutorials, assignments, and quizzes accessible 24/7 to enrolled students.
4. **Enable Online Assessment:** Allow teachers to create and administer quizzes with multiple question types (MCQ, True/False, Short Answer, Essay) and provide auto-grading for objective questions.
5. **Track Student Performance:** Generate subject-wise grade reports, GPA calculations, and progress visualizations to support data-driven academic interventions.
6. **Integrate Live Class Scheduling:** Connect students and teachers through Zoom-integrated virtual classrooms with automatic attendance marking.
7. **Provide Role-Based Access:** Enforce strict role-based permissions ensuring students, teachers, and administrators access only the data and functions relevant to their role.

8. **Generate Administrative Reports:** Produce enrollment, financial, performance, and audit reports exportable in PDF, Excel, and CSV formats.
9. **Ensure System Security:** Implement industry-standard security measures including bcrypt password hashing, AES-256 data encryption, CSRF protection, and comprehensive audit logging.

4. Scope

4.1. Included Features

The system encompasses the following functional boundaries:

- **Student Module:** Online registration, subject selection with automatic fee and discount calculation, learning material access, assignment submission, quiz attempts, grade and performance tracking, fee payment tracking, and live class participation via Zoom.
- **Teacher Module:** Course content management (upload/organize materials), assignment creation and management, online quiz creation with question banks, rubric-based grading with feedback, student performance monitoring, live class scheduling, and announcement broadcasting.
- **Administrator Module:** Full user management (students and teachers), academic program and subject management, class scheduling and timetable configuration, fee structure and discount policy management, financial reporting, faculty profile management, system configuration, backup and recovery management, and audit trail access.
- **System-Wide Features:** Role-based access control (RBAC), multi-channel notification system (email and in-system), advanced search functionality, session management with role-specific timeouts, and comprehensive activity logging.

4.2. Excluded from Initial Release

The following are explicitly out of scope for the first release:

- AI-based grading or personalized recommendations
- Multi-language support (English only)
- Parent portal or parental control features
- Offline functionality

- Integration with external school management systems
- Real-time in-platform video conferencing (Zoom is used externally)
- Social networking features among students

5. Data Collection

The data requirements and collection strategies for AAMS are described below.

5.1. Primary Sources

- **Requirement Provider Interviews:** Direct meetings with Mr. Ammar Ahmad Khan (Academy Owner) to elicit functional requirements, existing pain points, and operational workflows. Meeting minutes were documented and signed off.
- **Current Process Analysis:** Observation and documentation of existing manual processes including paper enrollment forms, handwritten fee ledgers, and WhatsApp-based assignment distribution.
- **User Surveys:** Structured questionnaires administered to a sample of students and teachers to understand usability expectations, device preferences, and accessibility needs.

5.2. Secondary Sources

- **IEEE Std 830-1998:** Standard for Software Requirements Specifications used as the structural and quality baseline for this document.
- **Existing SRS Documentation:** Previous version of the AAK Academy SRS reviewed to identify gaps addressed in this version.
- **Technical Documentation:** Official documentation for MySQL, HTML5, CSS3, JavaScript ES6+, and Zoom REST API reviewed to establish feasibility of proposed integrations.
- **Academic Regulations:** Punjab Board Matric and Intermediate program requirements reviewed to ensure the system aligns with academic calendar and grading standards.

5.3. Data Entities Managed by the System

Entity	Key Data Attributes
Student	Name, DOB, program, enrolled subjects, grades, fee status
Teacher	Name, qualifications, subject assignments, work-load
Subject	Title, fee, credit hours, program, prerequisites
Assignment	Instructions, deadline, submissions, grades, feedback
Quiz	Questions, time limit, scores, attempt history
Fee Record	Invoice, discounts applied, payments, balance
Attendance	Student/teacher presence per session, Zoom join logs
Audit Log	User action, timestamp, IP address, before/after values

6. Technology Stack

In accordance with the course lab structure, the system will utilize:

- **DBMS:** MySQL 8.0+ / MariaDB 10.5+ (Standard Relational DBMS).
- **Design Tools:** MySQL Workbench for EER Modeling and Schema Forward Engineering; Draw.io for system-level UML diagrams.
- **Query Language:** SQL (DDL, DML, DQL) for all transactional and analytical logic.
- **Frontend:** HTML5, CSS3, JavaScript (ES6+) with responsive design for cross-device compatibility.
- **Backend Integration:** PHP or Node.js for database connectivity, session management, and server-side business logic.
- **Version Control:** Git with GitHub (feature branching workflow).
- **Development Environment:** XAMPP / WAMP (local stack); Visual Studio Code (IDE).
- **API Integration:** Zoom REST API v2 (OAuth 2.0) for live class link generation and attendance sync.

7. System Functionality

The following are the key functional requirements the system will provide to users:

- F1. User Registration and Authentication:** Students self-register with personal, academic, and program details. Teachers are registered by the administrator. All users authenticate via email and password with role detection and secure session management.
- F2. Role-Based Access Control (RBAC):** The system enforces distinct permission sets for Students, Teachers, and Administrators, ensuring each user accesses only functions and data appropriate to their role.
- F3. Subject Selection with Real-Time Fee Calculation:** Students select subjects from available options (3–7 for Matric; 3–6 for Intermediate). The system calculates total fees in real-time and automatically applies a 10% discount for 3 subjects and a 20% discount for 6 subjects.
- F4. Learning Material Repository:** Teachers upload and organize lecture notes, presentations, video tutorials, practice exercises, and past papers. Students access and download materials organized by subject and topic.
- F5. Assignment Management:** Teachers create assignments with instructions, deadlines, and supporting files. Students upload completed work in multiple file formats. Teachers grade submissions and provide written feedback accessible to students after evaluation.
- F6. Online Quiz System:** Teachers create quizzes with MCQ, True/False, Short Answer, Essay, Matching, and Fill-in-the-Blank question types. Quizzes have configurable time limits. Objective questions are auto-graded; subjective questions require manual teacher review.
- F7. Grade and Performance Tracking:** Students view grades per assignment, quiz, and exam with subject-wise breakdowns, GPA calculations, progress charts, and comparison against class averages. Teachers identify at-risk students from class-wide distributions.
- F8. Live Class Integration (Zoom):** Teachers schedule Zoom meetings linked to specific subjects. Students join via one-click dashboard links. The system marks attendance automatically based on session participation.
- F9. Fee Payment and Invoice Management:** Students view fee invoices, payment history, applied discounts, and outstanding balances. Automatic reminders are sent before due dates; downloadable receipts are generated upon payment.

- F10. Administrator Dashboard and User Management:** Admins view system-wide metrics including enrollment statistics, financial summaries, and activity alerts. They approve or reject student registrations, manage teacher accounts, and assign subjects to teachers.
- F11. Academic Program and Subject Management:** Admins configure Matric and Intermediate programs, define subject fees, credit hours, and prerequisites, manage academic calendars, and set grading scales.
- F12. Faculty Information Page:** Public-facing teacher profiles display qualifications, experience, subject expertise, and contact information, organized by department for prospective students.
- F13. Report Generation:** The system produces enrollment, performance, financial, attendance, and audit reports filterable by date range, program, or user, exportable in PDF, Excel, and CSV formats.
- F14. Notification System:** Email and in-system notifications are triggered for assignment deadlines, quiz availability, grade releases, fee due dates, new announcements, and system alerts with customizable user preferences.
- F15. Audit Trail and Logging:** All user actions (login/logout, data changes, grade modifications, financial transactions, configuration changes) are logged with timestamp, user identity, and IP address for full accountability.
- F16. Backup and Recovery:** Scheduled automatic and manual database backups with encrypted off-site storage, integrity verification, and a documented recovery procedure targeting a Recovery Time Objective (RTO) of 24 hours and a Recovery Point Objective (RPO) of 1 hour.

8. User Classes

The system defines three primary user classes with distinct roles and access privileges:

User Class	Role Description	Access Level
Student	Matric or Intermediate enrolled learner. Uses the platform daily to access materials, submit assignments, attempt quizzes, and track academic progress.	Personal dashboard, enrolled subject content, own submissions, own quiz attempts, own grades, fee records, Zoom class links.
Teacher (Faculty)	Subject matter expert responsible for course delivery and student assessment. Manages content, creates assessments, grades work, and monitors performance.	Assigned course content management, assignment and quiz creation and grading, student performance for assigned classes, live class scheduling, announcements.
Administrator	Academy staff with full system oversight. Manages all users, academic programs, financial records, and system configuration.	Full system access: user management, program configuration, fee and discount management, report generation, system settings, audit logs, backup management.

Technical Proficiency by Role:

- **Students:** Basic web browsing; ability to fill forms, upload files, and use email. No programming knowledge required.
- **Teachers:** Moderate computer skills; familiarity with content upload, Zoom, and online assessment tools.
- **Administrators:** Intermediate computer skills; understanding of user management, database reporting, and system configuration.

9. Task Division and Collaboration

The project tasks are distributed to ensure individual accountability as per the course policy:

9.1. Salman Safdar (Group Lead)

- Implementation of DDL scripts (table creation, constraints, indexes, foreign keys).

- Development of complex DQL scripts involving multi-table joins, nested subqueries, and aggregate analytics for the fee and performance modules.
- Implementation of database triggers for automated audit logging and discount calculation enforcement.
- Backend API integration with Zoom REST API v2 for live class scheduling and attendance synchronization.

9.2. Muhammad Haris

- Design and implementation of the full frontend layer (HTML5/CSS3/ JavaScript) for all three user roles: Student, Teacher, and Administrator.
- Development of responsive dashboards, quiz interface with countdown timer, assignment submission UI with drag-and-drop, and grade visualization charts.
- Ensuring WCAG 2.1 Level AA accessibility compliance across all interface components.
- Integration of frontend forms with backend server-side endpoints for real-time fee calculation and subject selection.

9.3. Farwa Imran

- Design of the comprehensive EER diagram capturing all entities, relationships, cardinalities, and participation constraints.
- Conversion of EER diagrams into relational schemas and analysis of functional dependencies.
- Performing normalization up to 3NF to ensure data integrity and remove insertion, deletion, and update anomalies.
- Requirements elicitation, SRS documentation, test case authoring, and coordination with the requirement provider (Mr. Ammar Ahmad Khan) for validation sign-off.

10. Conclusion

The AAK Academy Management System represents a comprehensive and targeted digital transformation solution for AAK Academy. By replacing fragmented, manual processes with an integrated web-based platform, the system directly addresses the operational inefficiencies, financial inaccuracies, and academic transparency gaps identified during requirements gathering.

The proposed system delivers measurable value across all stakeholder groups. Students gain 24/7 access to learning materials, transparent grade tracking, and a streamlined enrollment experience. Teachers benefit from powerful content management and assessment tools that reduce administrative burden. Administrators receive a centralized command center with real-time analytics, automated reporting, and complete financial oversight.

The automated discount system, Zoom-integrated live classes, role-based dashboards, and comprehensive audit logging together form a platform that is not only functionally robust but also secure, scalable, and maintainable. Built on open-source technologies (HTML5, CSS3, JavaScript, MySQL), the system minimizes licensing costs while ensuring long-term viability and sustainability.

Upon successful deployment, AAMS is expected to reduce administrative workload by approximately 70%, eliminate fee calculation errors, and significantly improve student engagement and academic performance visibility. This project demonstrates the power of thoughtfully engineered software to transform educational institutions and serves as a replicable model for similar academies across Pakistan.

AI Usage Disclosure

- **AI Tool:** Claude (Anthropic).
- **Purpose:** Generating L^AT_EX boilerplate, aligning SRS deliverables with CSC-225 CLOs, expanding technical descriptions, and structuring the relational schema.
- **Access Date:** February 2026.